Robust Classical Machine Learning for Biomedical Diagnosis: A Dual Evaluation on Breast Cancer Prediction and Ear-EEG Emotion Recognition

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Abstract

Reliable analysis of biomedical data is essential for diagnostic support and human-state monitoring. This study investigates the performance and robustness of classical machine-learning models across two different biomedical tasks: breast-cancer classification using the Wisconsin Diagnostic dataset and emotion recognition using in-ear EEG signals. For the cancer dataset, four traditional classifiers were evaluated under clean conditions, reduced training sets, and multiple noise intensities. The results show that KNN provides the highest stability, while Naive Bayes is strongly affected by severe noise, indicating fundamental sensitivity to distributional perturbations.

For the EEG task, handcrafted temporal–spectral features were extracted from two channels and combined with Fisher Score and Mutual Information for feature selection. A grid-search procedure optimized each classifier, and an enhanced feature set was developed by incorporating db4 wavelet-based descriptors. This augmentation substantially increased performance, with SVM showing the largest improvement, raising its mean F1-score from about 0.67 to **0.79**.

Overall, the findings demonstrate that classical machine-learning algorithms remain effective for structured clinical data and low-channel EEG recordings. The wavelet-enhanced feature extraction significantly strengthens EEG emotion classification, highlighting its suitability for practical, lightweight biomedical systems.

Keywords

Classical Machine Learning; Biomedical Signal Processing; Breast Cancer Classification; Ear-EEG Emotion Recognition; Supervised Classification.

Introductio

Machine learning has become an essential component of modern biomedical analysis, supporting diagnostic decision-making and enabling automated interpretation of complex biosignals. Classical machine-learning algorithms remain highly relevant due to their interpretability, computational efficiency, and strong performance in scenarios where dataset size is limited or the underlying feature space is well structured. These characteristics make traditional models particularly suitable for medical applications in which transparency, reliability, and ease of deployment are critical [1].

The first part of this study investigates breast cancer prediction using the Breast Cancer Wisconsin Diagnostic (WDBC) dataset, a widely used benchmark composed of handcrafted morphological features extracted from fine-needle aspirate images. Prior research shows that classical classifiers such as K-Nearest Neighbors, Support Vector Machines, and Naive Bayes can achieve competitive diagnostic accuracy on this dataset, often rivaling more complex deep-learning architectures when the amount of available data is modest [2]. Furthermore, evaluating model behavior under reduced training data and artificially injected noise is essential, as clinical measurements frequently exhibit acquisition variability, missing data, or sample imbalance.

The second part of this work focuses on affective-state classification using compact in-ear EEG recordings. Recent developments in ear-EEG technology have introduced lightweight, unobtrusive systems capable of capturing neural activity in real-world environments. Emotion recognition from EEG commonly relies on handcrafted temporal and spectral features combined with classical learning algorithms, which often generalize more reliably than deep neural networks on small or noisy EEG datasets [3]–[4]. These approaches remain important for mobile and wearable neurotechnology, where data non-stationarity and subject variability present persistent challenges.

Although the two domains differ in signal nature—static cytological measurements versus non-stationary electrophysiological activity—they share a unified data-analysis pipeline consisting of preprocessing, feature extraction, feature selection, and classification. By jointly presenting results across these heterogeneous datasets, this work highlights the versatility and robustness of classical machine-learning models and demonstrates how feature engineering and dimensionality reduction influence classifier performance under realistic biomedical constraints.

Methods

2.Dataset Description

A. Breast Cancer Wisconsin Diagnostic (WDBC) Dataset

The first dataset employed in this study is the Breast Cancer Wisconsin Diagnostic (WDBC) dataset, a well-established benchmark for evaluating classical machine-learning models in medical diagnosis. The dataset contains 569 patient samples, each derived from a fine-needle aspirate (FNA) of a breast mass. Every sample includes 30 continuous-valued diagnostic features characterizing nuclear morphology, including radius, texture, perimeter, area, concavity, compactness, smoothness, symmetry, and several “worst” statistical descriptors computed over multiple cell nuclei.
Each instance is assigned one of two diagnostic categories:

- 0 — Malignant
- 1 — Benign

The dataset is clean, preprocessed, and widely recognized for its structured feature space with moderate class separability. In this study, samples were partitioned into 70% training (398 samples) and 30% testing (171 samples). Standard normalization was applied to all features prior to model training. No additional preprocessing was required because all 30 diagnostic attributes are provided in ready-to-use format.

B. In-Ear EEG Emotional State Dataset

The second dataset comprises two-channel in-ear EEG recordings obtained using a miniature custom-built system with electrodes positioned inside the left and right ear canals. This configuration offers stable, noise-resistant recording suitable for emotion recognition experiments.

1) Experimental Protocol

Participants completed 21 video–emotion trials, Each trial therefore produced approximately 10 seconds of EEG data corresponding to the emotional stimulus, followed by a valence rating.

2) Signal Acquisition and Segmentation

EEG data were recorded at a sampling frequency of 500 Hz.
From each 10-second emotional segment:

- Signals were divided into 2.5-second windows
- with 50% temporal overlap
- producing windows of 1250 samples per channel
- for both left and right in-ear electrodes

This windowing procedure increases sample count while preserving temporal information essential for emotion classification.

3) Preprocessing Pipeline

All extracted EEG windows were filtered within the 4–45 Hz band to remove low-frequency drift and high-frequency noise. Subsequently, each window was decomposed into the five standard EEG frequency ranges:

- δ (1–4 Hz)
- θ (4–8 Hz)
- α (8–12 Hz)
- β (12–30 Hz)
- γ (30–45 Hz)

These band-limited signals were used for feature extraction.

4) Label Assignment

Each EEG window inherited the valence label associated with its corresponding trial:

- +1 — Positive
- 0 — Neutral
- –1 — Negative

Because emotional responses are not uniformly distributed across classes, the dataset exhibits natural class imbalance, and balancing strategies were applied where required.

3. Feature Extraction

Feature extraction was performed separately for the two datasets used in this study:
(1) the Breast Cancer Wisconsin dataset, which already provides 30 descriptive clinical features per sample and
(2) the two-channel in-ear EEG dataset, for which all features were computed manually from the raw signals.
This section describes the complete extraction pipeline for both datasets.

3.1 Breast Cancer Dataset Features

The Breast Cancer Wisconsin dataset contains 30 handcrafted morphological features extracted from digitized images of fine-needle aspirates. These features describe cell shape, texture, radius, smoothness, concavity, compactness, area, symmetry, and several error metrics. Because these attributes are already computed in the original dataset, no additional feature construction was required for this part of the study.

3.2 EEG Feature Extraction Pipeline

For the EEG dataset, a comprehensive feature extraction process was applied to each 2.5-second segment from both channels. The goal was to capture both temporal and spectral properties of the in-ear EEG signals across the five major frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–45 Hz).
For each channel and each frequency band, eight temporal–spectral features were computed, resulting in:

- 8 features × 5 bands × 2 channels = 80 band-specific temporal features
- 20 wavelet-based features (10 per channel)
- Total per sample: 100 features

The extracted features are detailed below.

3.2.1 Band-Specific Temporal Features

For each frequency-filtered EEG signal segment x(t) of length N samples, the following features were computed.

1) Mean (μ)

$$\mu = (1/N) \cdot \Sigma\, x(t)$$

2) Standard Deviation (σ)

$$\sigma = sqrt(\,(1/N) \cdot \Sigma\,(\,x(t) - \mu\,)^2\,)$$

3) Mean of First-Order Differences (D_1)

$$D_1 = (1/(N-1)) \cdot \Sigma\,|\,x(t+1) - x(t)\,|$$

4) Normalized First-Order Differences (D_1_norm)

First normalize the signal:

$$x\_norm(t) = (\,x(t) - \mu\,)\,/\,\sigma$$

Then:

$$D_1\_norm = (1/(N-1)) \cdot \Sigma\,|\,x\_norm(t+1) - x\_norm(t)\,|$$

5) Mean of Second-Order Differences (D_2)

$$D_2 = (1/(N-2)) \cdot \Sigma\,|\,x(t+2) - x(t)\,|$$

6) Normalized Second-Order Differences (D_2_norm)

$$D_2\_norm = (1/(N-2)) \cdot \Sigma\,|\,x\_norm(t+2) - x\_norm(t)\,|$$

7) Relative Power Spectral Density (rPSD)

Computed using the band-limited power divided by total broadband power:

$$rPSD = (\,\int P\_x(f)\,df\,)\,/\,(\,\int P\_total(f)\,df\,)$$

8) Differential Entropy (DE)

Assuming a Gaussian distribution:

$$DE = 0.5 \cdot log(\,2\pi e \cdot \sigma^2\,)$$

Differential entropy is widely used in EEG emotion-recognition literature due to its stability and discriminative power in short window segments.

3.2.2 Wavelet-Based Features

In addition to the band features above, wavelet features were computed directly from the raw EEG signals.
For each channel, a discrete wavelet transform (Daubechies-4, level = 4) was applied. From the resulting detail and approximation coefficients, ten statistical descriptors were extracted (e.g. energy, entropy), yielding:

- 10 wavelet features × 2 channels = 20 wavelet features

These wavelet descriptors complement the band-power-based features by capturing transient and multi-resolution dynamics in the in-ear EEG.

3.2.3 Final Feature Vector

Combining all components:

- 80 temporal–spectral features
- 20 wavelet features

Each EEG sample is represented by a 100-dimensional feature vector, suitable for classical machine learning classifiers.

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EEG Data Balancing

The EEG dataset contained 56 samples of label +1, 56 samples of label –1, and 28 samples of label 0. To balance the classes, the minority class (0) was oversampled using a Mixup-based method: [5]

$$x\_new = \lambda \cdot x_1 + (1 - \lambda) \cdot x_2$$

With $\lambda = 0.4$.

Synthetic samples were generated until the neutral class reached 56 samples, resulting in a balanced distribution of:

- 56 positive (+1)
- 56 negative (–1)
- 56 neutral (0)

Feature Reduction

Two filter-based feature selection methods were applied to reduce the dimensionality of the EEG feature space: Fisher Score (ANOVA F-test) and Mutual Information (MI). Both methods evaluate each feature independently and retain the top-k most informative features.

1) Fisher Score (ANOVA F-Test)

The Fisher criterion measures how well a feature separates class means relative to within-class variance. For a feature (x) across two classes, the Fisher score is:

$$F = ((\mu_1 - \mu_2)^2) / (\sigma_1{}^2 + \sigma_2{}^2)$$

where

μ_1, μ_2 = mean of x in class 1 and class 2
σ_1^2, σ_2^2 = variance of x in class 1 and class 2

In multi-class settings, the ANOVA F-test generalizes this ratio using between-class and within-class variances. Higher F-values correspond to stronger discrimination.

2) Mutual Information (MI)

Mutual Information quantifies the dependency between a feature (x) and the class label (y). It is defined as:

$$MI(x, y) = \Sigma\, \Sigma\, p(x, y) \cdot log(\, p(x, y) \,/\, (\, p(x) \cdot p(y)\,)\,)$$

where

p(x, y) = joint probability of x and y
p(x), p(y) = marginal probabilities

Features with larger MI values provide more information about the class labels and capture nonlinear relationships. [6]

Selection Strategy

Both Fisher and MI scores were computed for all features, ranked in descending order, and the top-k features were selected using the SelectKBest procedure.

4. Classification Methods

This study employed different classical classifiers for the two datasets. Distance-based and probabilistic models were used for the Breast Cancer dataset, while linear discriminative models were applied to the EEG dataset due to the higher dimensionality and continuous nature of the features.

A. Classifiers for the Breast Cancer Dataset [7]

1) K-Nearest Neighbors (KNN)

KNN assigns a test sample to the majority class among its k nearest neighbors according to the Euclidean distance:

$$d(x, x_k) = sqrt(\,\Sigma\,(\,x_j - x_{kj}\,)^2\,)$$

Prediction is obtained by:

$$\hat{y} = mode(\,y_1, y_2, \ldots, y_k\,)$$

2) Nearest Mean Classifier (NMC)

Each class c is represented by its centroid:

$$\mu\_c = (1 / N\_c) \cdot \Sigma x_i$$

A test sample is assigned to the nearest class mean:

$$\hat{y} = argmin\_c \, || x - \mu\_c ||$$

3) Naive Bayes (NB)

Based on Bayes' rule with conditional independence:

$$P(c \mid x) \propto P(c) \cdot \Pi P(x_j \mid c)$$

For Gaussian NB:

$$P(x_j \mid c) = (1 / (sqrt(2\pi) \, \sigma\_c)) \cdot exp(-(x_j - \mu\_c)^2 / (2\sigma\_c^2))$$

Prediction is:

$$\hat{y} = argmax\_c \, P(c \mid x)$$

4) Parzen Window Classifier

Class-conditional densities are estimated using kernel functions:

$$p(x \mid c) = (1 / (N\_c \, h^D)) \cdot \Sigma K((x - x_i) / h)$$

Classification uses:

$$\hat{y} = argmax\_c \, P(c) \cdot p(x \mid c)$$

B. Classifiers for the EEG Dataset [8]

1) Support Vector Machine (SVM)

SVM finds a maximum-margin separating hyperplane:

$$f(x) = sign(w^T x + b)$$

with optimization:

$$min \, (1/2) \, ||w||^2$$
$$subject \; to \; y_i \, (w^T x_i + b) \geq 1$$

Kernels (e.g., RBF) allow nonlinear decision boundaries.

2) Linear Discriminant Analysis (LDA)

LDA maximizes between-class variance relative to within-class variance:

$$w = S\_w^{-1} (\mu_1 - \mu_2)$$

with decision:

$$\hat{y} = sign(w^T x + b)$$

3) Logistic Regression (LR)

Posterior probability:

$$P(y = 1 \mid x) = 1 / (1 + exp(-(w^T x + b)))$$

Parameters are learned via cross-entropy minimization:

$$L = -\Sigma [\, y_i \, log(p_i) + (1 - y_i) \, log(1 - p_i) \,]$$

Results

Breast Cancer Dataset

A. Scatter-Plot Visualization of Feature Pairs

To examine the discriminative structure of the WDBC dataset, three pairs of features were visualized using scatter plots. The dataset was divided into 70% training and 30% testing, and the plots were generated using the test set. The visualizations illustrate how different feature combinations separate malignant and benign samples:

1. Mean Radius vs. Worst Fractal Dimension
   Malignant samples occupy higher-value regions along both axes, indicating strong class separation.
2. Worst Concavity vs. Concavity Error
   This pair exhibits clear divergence between classes, with malignant samples showing larger concavity and error values.
3. Worst Smoothness vs. Compactness Error
   Although the separation is more moderate, malignant samples still cluster toward higher compactness-error regions.

These plots demonstrate that several feature pairs provide meaningful separation between the two classes, supporting the suitability of classical classifiers for this dataset.

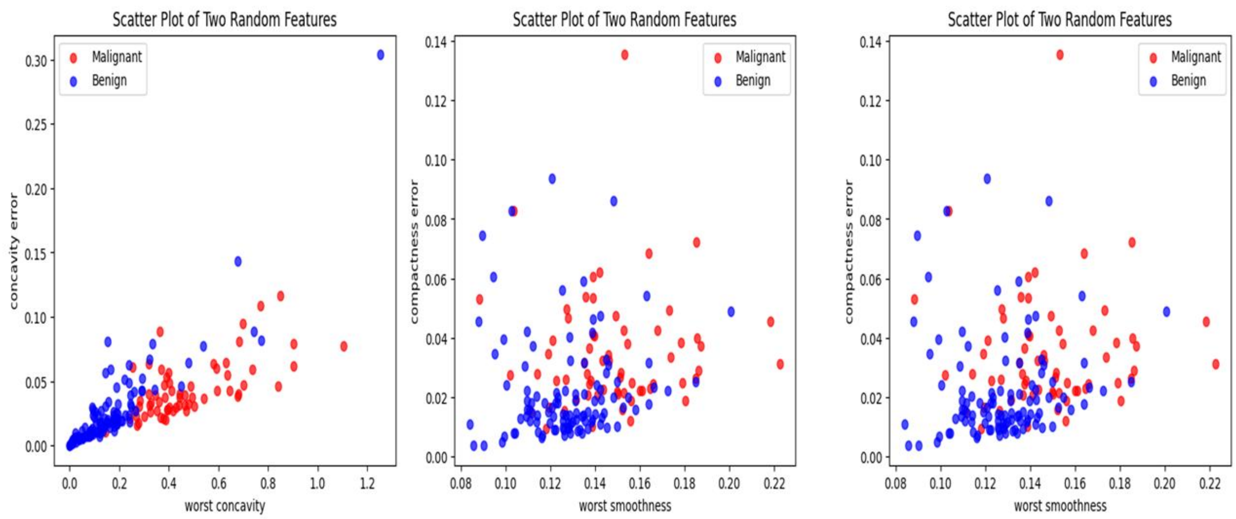

Figure 1. Scatter-plot visualization of three diagnostic feature pairs from the WDBC testset:

B. Classification Results on the WDBC Dataset

The classification performance of four classical models—K-Nearest Neighbors (KNN), Nearest Mean Classifier (NMC), Naive Bayes (NB), and Parzen Window Classifier (PWC)—was evaluated on the WDBC dataset. Models were trained using 70% of the samples and tested on the remaining 30%. Table 1 summarizes the achieved accuracy and F1-scores, followed by the confusion matrices for each classifier.

Table 1. Classification Performance on the WDBC Test Set

Classifier	**Accuracy**	**F1-Score**
K-Nearest Neighbors (KNN)	**0.9883**	**0.9907**
Nearest Mean Classifier	0.9591	0.9674
Naive Bayes	0.9474	0.9573
Parzen Window Classifier	0.9591	0.9677

Confusion Matrices

1) K-Nearest Neighbors (KNN)

```
                Predicted
              Malignant   Benign
Actual
Malignant         62         2
Benign             0       107
```

2) Nearest Mean Classifier (NMC)

```
                Predicted
              Malignant   Benign
Actual
Malignant         60         4
Benign             3       104
```

3) Naive Bayes (NB)

```
                Predicted
              Malignant   Benign
Actual
Malignant         61         3
Benign             6       101
```

4) Parzen Window Classifier (PWC)

```
                Predicted
              Malignant   Benign
Actual
Malignant         59         5
Benign             2       105
```

C. Effect of Training Set Size on Classifier Performance

To evaluate how the number of available training samples affects model reliability, the WDBC dataset was subsampled to 20%, 40%, 60%, and 80% of its original size. Four classical classifiers (K-Nearest Neighbors, Nearest Mean Classifier, Naive Bayes, and Parzen Window Classifier) were trained on each subset and evaluated on the same fixed test set. Accuracy and F1-score were reported for all configurations.

Numerical Results

Table 2. Classification performance under varying training set sizes

Training Size	Classifier	Accuracy	F1-score
20%	KNN	0.919	0.937
	Nearest Mean	0.919	0.936
	Naive Bayes	0.925	0.940
	Parzen Window	0.921	0.937
40%	KNN	0.959	0.968
	Nearest Mean	0.936	0.950
	Naive Bayes	0.933	0.947
	Parzen Window	0.965	0.972
60%	KNN	0.974	0.979
	Nearest Mean	0.947	0.959
	Naive Bayes	0.943	0.955
	Parzen Window	0.947	0.958
80%	KNN	0.991	0.993
	Nearest Mean	0.956	0.966
	Naive Bayes	0.965	0.973
	Parzen Window	0.974	0.979

Interpretation

Reducing the size of the training set affects the four classifiers differently. KNN shows the largest performance drop at low training sizes because its neighborhood-based decisions become unreliable when only a small portion of the feature space is populated.

Nearest Mean is less sensitive but still impacted, since poorly estimated class centroids lead to biased decisions. Naive Bayes remains relatively stable because its parametric assumptions allow it to generalize even with limited data, although small-sample parameter estimates still reduce accuracy.

Parzen Window is highly dependent on sample density, so limited data leads to noisy kernel estimates, but its performance improves sharply as more training data become available.

Overall, non-parametric and instance-based models benefit most from larger training sets, while parametric models degrade more gradually under data scarcity.

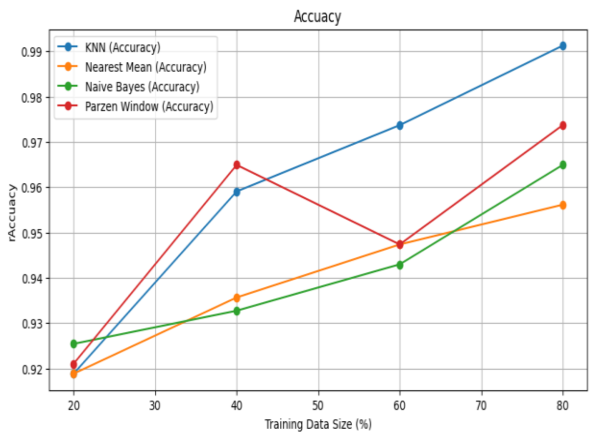

Figure 2. Accuracy of four classifiers across different training set sizes.

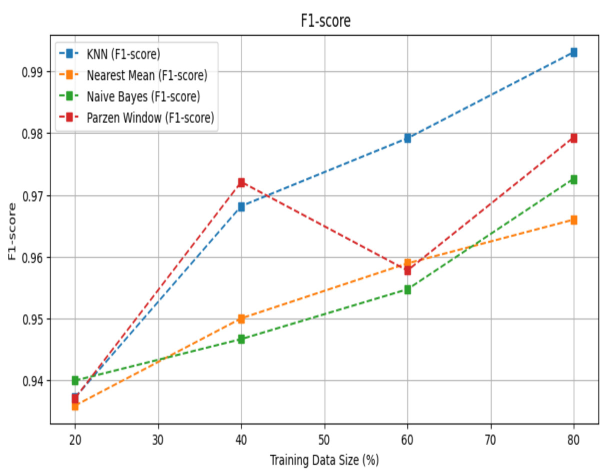

Figure 3. F1-score of four classifiers across different training set sizes.

D. Noise Robustness Analysis

To assess the robustness of the classifiers under test-time perturbations, two levels of Gaussian noise were added *only to the test samples*:
(1) **small noise** with low variance, and
(2) **large noise** with higher variance.
No noise was added to the training data. All four classifiers were evaluated on clean, small-noise, and large-noise test sets.

Numerical Results

Table 3. Performance of classifiers under different noise conditions

Classifier	**Acc (Clean)**	**F1 (Clean)**	**Acc (Small Noise)**	**F1 (Small Noise)**	**Acc (Large Noise)**	**F1 (Large Noise)**
KNN	0.988	0.991	0.977	0.981	0.936	0.949
Nearest Mean	0.959	0.967	0.959	0.967	0.924	0.940
Naive Bayes	0.947	0.957	0.912	0.927	0.409	0.106
Parzen Window	0.959	0.968	0.959	0.967	0.883	0.907

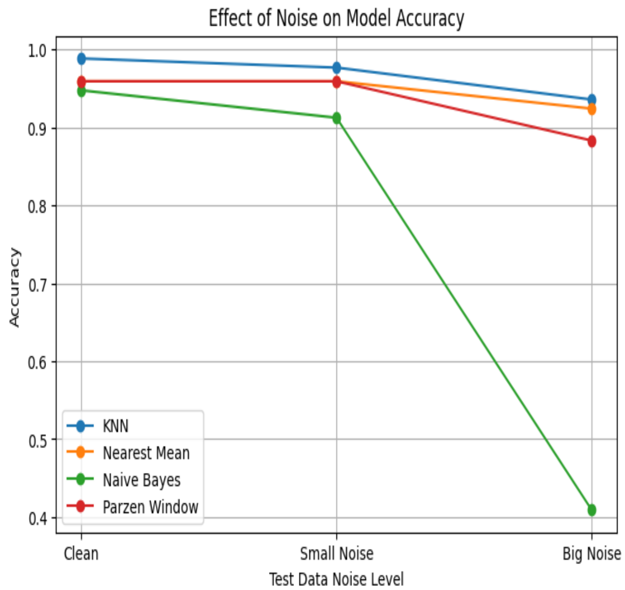

Figure 4. Accuracy degradation of four classifiers under clean, small-noise, and large-noise test conditions.

Interpretation

Small noise causes only minor changes in model performance. KNN and Nearest Mean remain stable because neighborhood structure and class means are only slightly perturbed. Parzen Window shows a small drop for the same reason, as kernel density estimates stay consistent under light distortion. In contrast, large noise produces stronger degradation. KNN and Nearest Mean decline moderately as distances and class boundaries become less reliable. Parzen Window is affected more noticeably because noise spreads samples in feature space, reducing density precision. Naive Bayes is impacted the most: noise disrupts its distributional assumptions, causing a sharp drop in performance under high-noise conditions.

Classification Performance on the EEG Dataset

Two feature-selection methods (Fisher score and Mutual Information) were combined with four classical classifiers—Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), Logistic Regression, and Decision Tree—resulting in eight classification pipelines. For each configuration, classification performance was evaluated using 5-fold cross-validation, and the mean F1-score with standard deviation was recorded.

For each of the eight configurations, a grid-search procedure was applied to jointly optimize the classifier hyperparameters and the number of selected features. The search was conducted within the cross-validation loop, ensuring that both the model parameters and the optimal value of k in the feature-selection stage were determined based exclusively on the training folds.

Table 4. Mean F1-score (5-fold) for all feature-selection/classifier combinations

Feature Method	Classifier	Mean F1	Std
Fisher	SVM	0.6747	0.0516
Fisher	LDA	0.5684	0.0475
Fisher	DecisionTree	0.4881	0.0819
Fisher	LogReg	0.6065	0.0785
MutualInfo	SVM	0.6022	0.0769
MutualInfo	LDA	0.5749	0.0802
MutualInfo	DecisionTree	0.4902	0.0942
MutualInfo	LogReg	0.6085	0.0730

Interpretation

The results highlight several consistent patterns across both feature-selection strategies. SVM achieves the highest F1-scores overall, particularly when combined with Fisher scoring. This suggests that nonlinear decision boundaries are beneficial for distinguishing emotional EEG segments, and that Fisher-ranked features retain structure that is useful for radial-basis SVM models. Logistic Regression performs competitively, especially under Mutual Information, indicating that a linear classifier can still generalize effectively when features capturing dependency information are selected.

LDA shows moderate performance. Since LDA assumes Gaussian class-conditional distributions with equal covariance, the variability and nonstationary nature of EEG signals likely limit its ability to model class boundaries accurately. Decision Trees exhibit the lowest F1-scores across all configurations. High-dimensional EEG features tend to lead to unstable splits and overfitting in tree-based models, which explains the larger standard deviations observed in cross-validation.

Comparing the feature-selection strategies, Fisher scoring consistently yields higher performance for SVM and LDA, suggesting that variance-based discrimination aligns more effectively with these classifiers. Mutual Information performs similarly for Logistic Regression and slightly worse for SVM, likely due to the sensitivity of SVM kernels to the distributional properties of selected features.Overall, the results indicate that (i) nonlinear models benefit most from Fisher-selected features, (ii) linear models remain relatively stable across both selection criteria, and (iii) tree models are not well suited for this type of EEG data

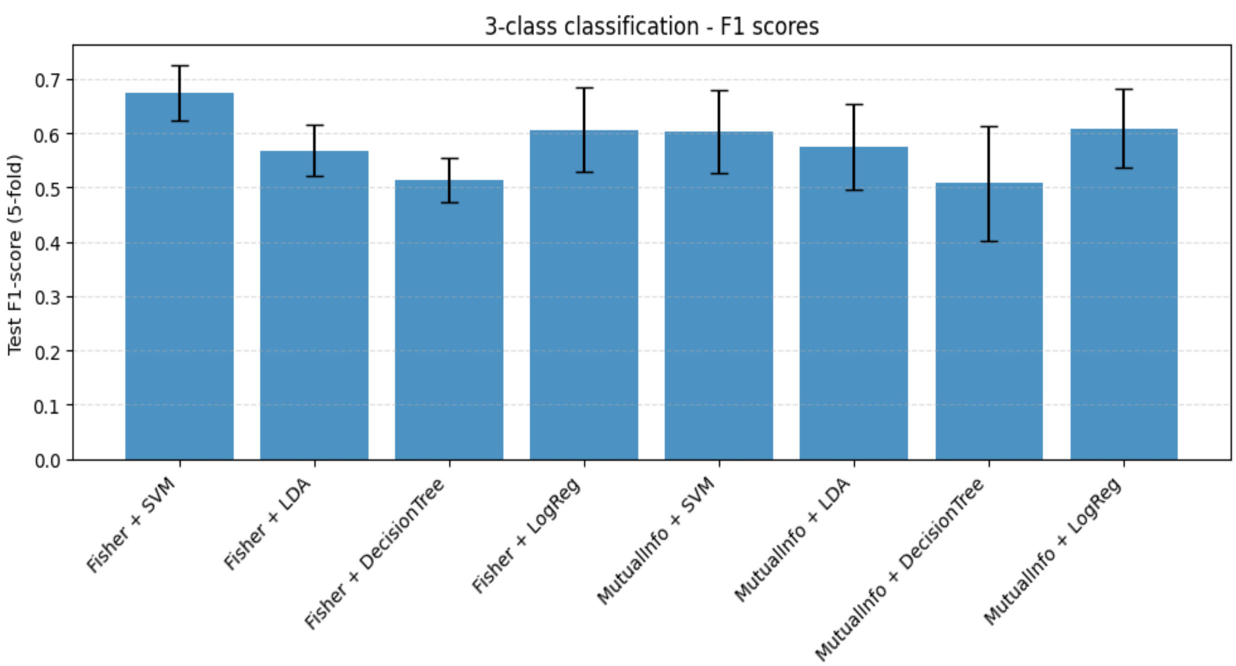

Figure 5. Mean 5-fold F1-scores for all classifier–feature selection combinations.

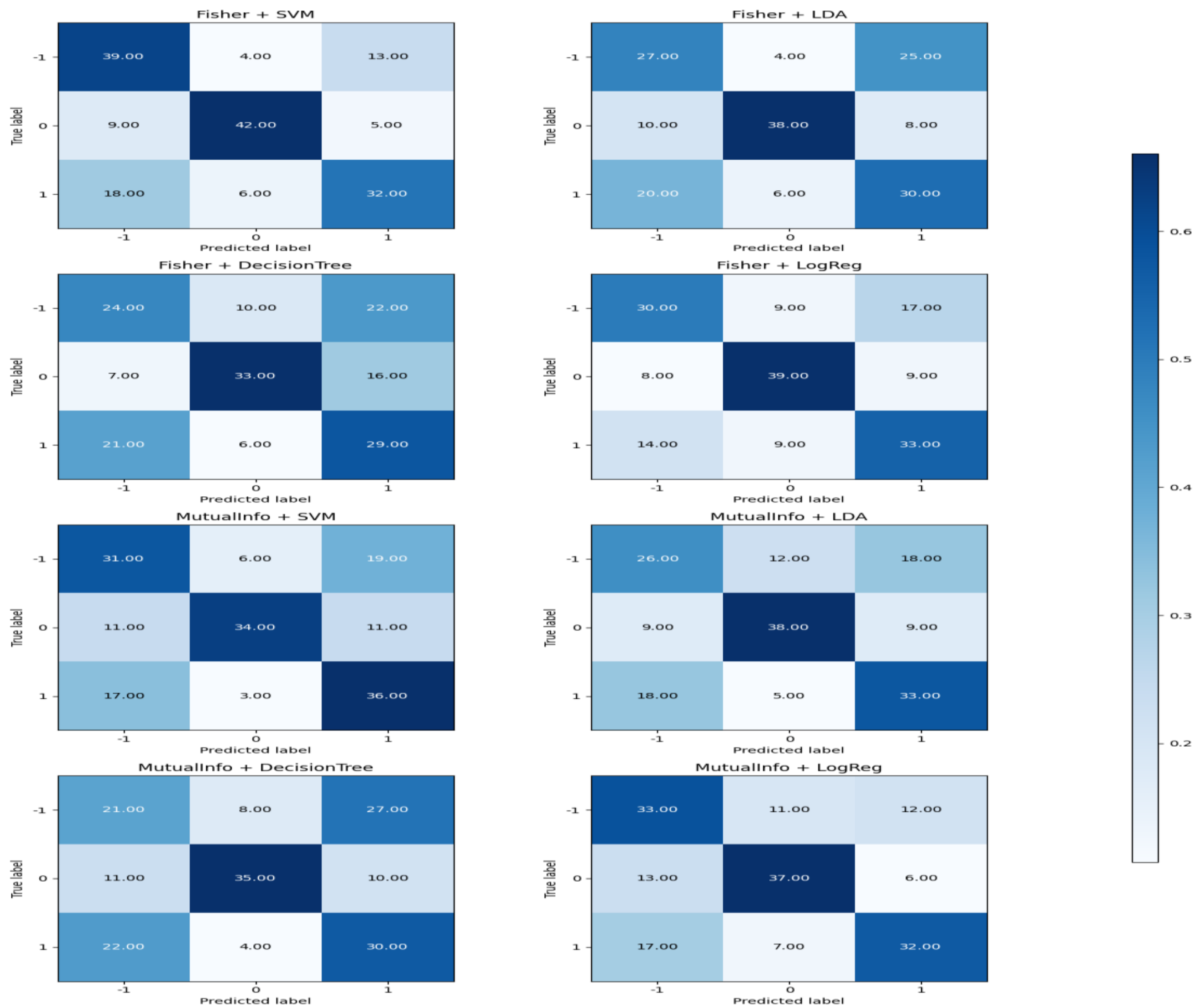

Fig. 6. Confusion matrices for the eight feature-selection/classifier combinations on the EEG datase

Proposed Method: EEG Classification with Extended Wavelet Features

In the proposed approach, the original 80 handcrafted EEG features were augmented with 20 additional wavelet-based descriptors extracted from both ear-canal channels using the db4 mother wavelet. These features capture localized temporal–frequency fluctuations that are not represented in the baseline feature set. After concatenation, the full feature vector was processed using the same two feature-selection criteria (Fisher score and Mutual Information) and classified using SVM, LDA, Decision Tree, and Logistic Regression. Five-fold cross-validation was applied to ensure robust evaluation.

1) Results with Fisher Feature Selection

Model	**Mean F1 (5-fold)**	**Std**
SVM	0.7912	0.0718
LDA	0.6804	0.0476
Decision Tree	0.5594	0.0721
Logistic Reg.	0.6204	0.1075

2) Results with Mutual Information

Model	Mean F1 (5-fold)	Std
SVM	0.6969	0.0661
LDA	0.5919	0.0593
Decision Tree	0.6138	0.1030
Logistic Reg.	0.6564	0.0313

3) Comparison with the Baseline Method

Across all classifiers and both feature-selection strategies, the inclusion of wavelet features substantially improved performance compared to the baseline. The most notable gains occurred for SVM and Logistic Regression, which both benefit from richer feature representations. Fisher-based feature selection combined with SVM achieved the highest overall performance, improving from approximately 0.67 F1 in the baseline to nearly 0.79 F1 in the proposed method. Decision Tree models also showed moderate improvement, though with larger variability.

4) Interpretation

The wavelet features enhanced the feature space by introducing discriminative temporal–frequency information that complements the original handcrafted descriptors. Classifiers that rely on smooth decision boundaries, such as SVM and Logistic Regression, benefited most because the added features reduce class overlap in the transformed space. Mutual Information and Fisher criteria effectively identified the most informative subset of the expanded 100-dimensional feature vector. Overall, the results demonstrate that augmenting EEG data with wavelet-based descriptors significantly improves model performance for three-class emotion classification.

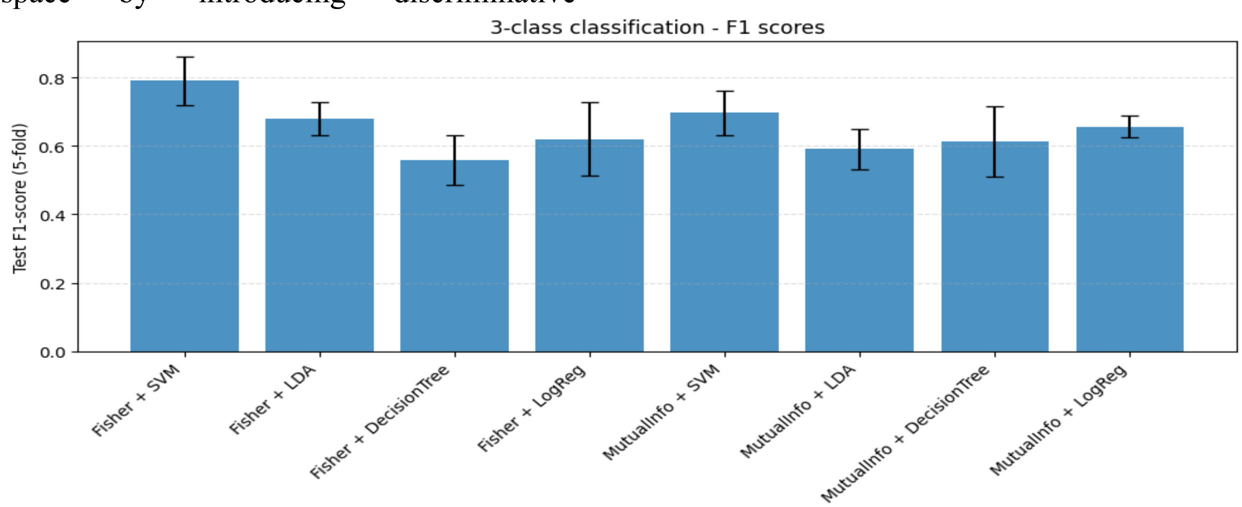

Figure 7. Mean 5-fold F1-scores for all classifier–feature selection combinations.

Fig. 8 . Confusion matrices for the eight feature-selection/classifier combinations on the EEG dataset

Binary classification analysis with the proposed EEG features

In addition to the 3-class setting, the proposed feature set (original 80 features + 20 db4 wavelet features from two channels) was evaluated on three binary problems: (i) Class 0 vs 1, (ii) Class 0 vs −1, and (iii) Class 1 vs −1. For each task, two feature selection methods (Fisher score and Mutual Information) and four classifiers (SVM, LDA, Decision Tree, Logistic Regression) were combined. Performance was measured using 5-fold cross-validated F1-score.

1) Class 0 vs 1

Table 5. F1-scores for Class 0 vs 1 (5-fold)

Feature selection	**Classifier**	**Mean F1**	**Std**
Fisher	SVM	0.8203	0.0558
	LDA	0.7278	0.0611
	Decision Tree	0.6588	0.0697
	Logistic Reg.	0.7222	0.0604
MutualInfo	SVM	0.7669	0.0478

Feature selection	Classifier	Mean F1	Std
	LDA	0.8306	0.0642
	Decision Tree	0.7562	0.0552
	Logistic Reg.	0.6839	0.0981

For discrimination between classes 0 and 1, the best performance is obtained with LDA + Mutual Information (F1 ≈ 0.83) and SVM + Fisher (F1 ≈ 0.82). Both methods clearly outperform Decision Tree and Logistic Regression. This shows that the added wavelet features give enough linear separability for LDA while still supporting a strong non-linear margin for SVM.

2) Class 0 vs −1

Table 6. F1-scores for Class 0 vs −1 (5-fold)

Feature selection	Classifier	Mean F1	Std
Fisher	SVM	0.7320	0.0677
	LDA	0.7924	0.0641
	Decision Tree	0.6499	0.0782
	Logistic Reg.	0.7093	0.0873
MutualInfo	SVM	0.7320	0.0677
	LDA	0.7859	0.0993
	Decision Tree	0.6997	0.0800
	Logistic Reg.	0.7183	0.0636

For the 0 vs −1 task, LDA again achieves the highest F1-scores (around 0.79–0.80), slightly better with Fisher than with Mutual Information. SVM and Logistic Regression form a second group with F1 around 0.71–0.73, while Decision Tree remains weaker. The results suggest that the feature space for these two classes is almost linearly separable, and that the db4 wavelet features help LDA capture this separation reliably.

3) Class 1 vs −1

Table 7. F1-scores for Class 1 vs −1 (5-fold)

Feature selection	Classifier	Mean F1	Std
Fisher	SVM	0.7203	0.0779
	LDA	0.6343	0.0707
	Decision Tree	0.5518	0.0955
	Logistic Reg.	0.7195	0.0954
MutualInfo	SVM	0.6829	0.0795
	LDA	0.6194	0.0601
	Decision Tree	0.6817	0.0400
	Logistic Reg.	0.7198	0.0933

The Class 1 vs −1 problem is more difficult, with F1-scores around 0.55–0.72. SVM and Logistic Regression give the best and most stable results (F1 ≈ 0.72) when combined with either Fisher or Mutual Information. LDA and Decision Tree perform clearly worse. Here the wavelet features still improve performance compared to the previous feature set, but the overlap between classes 1 and −1 remains higher than in the other binary tasks.

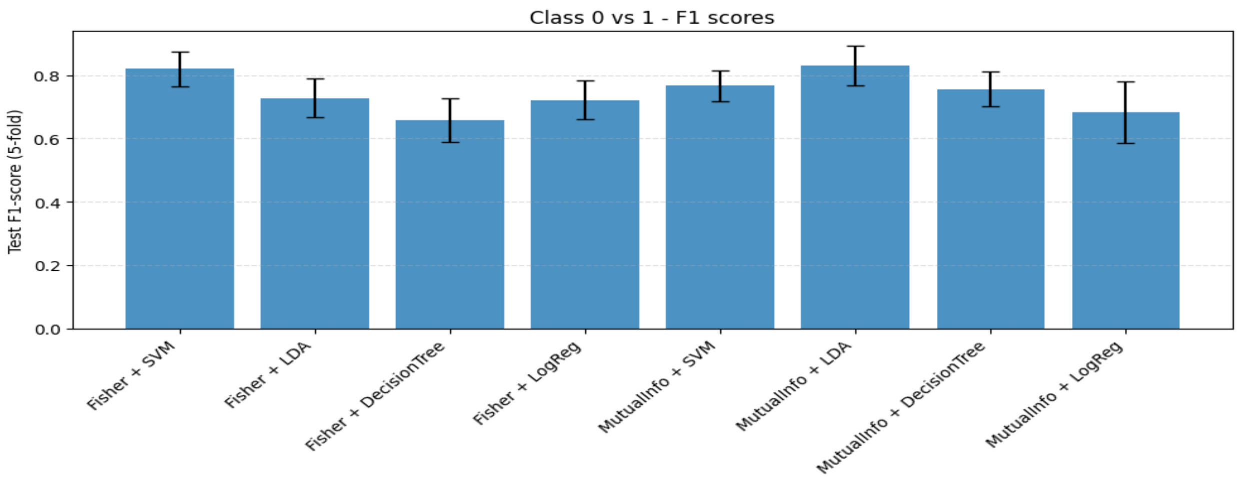

Figure 9.1 – F1-scores for binary classification (Class 0 vs 1).

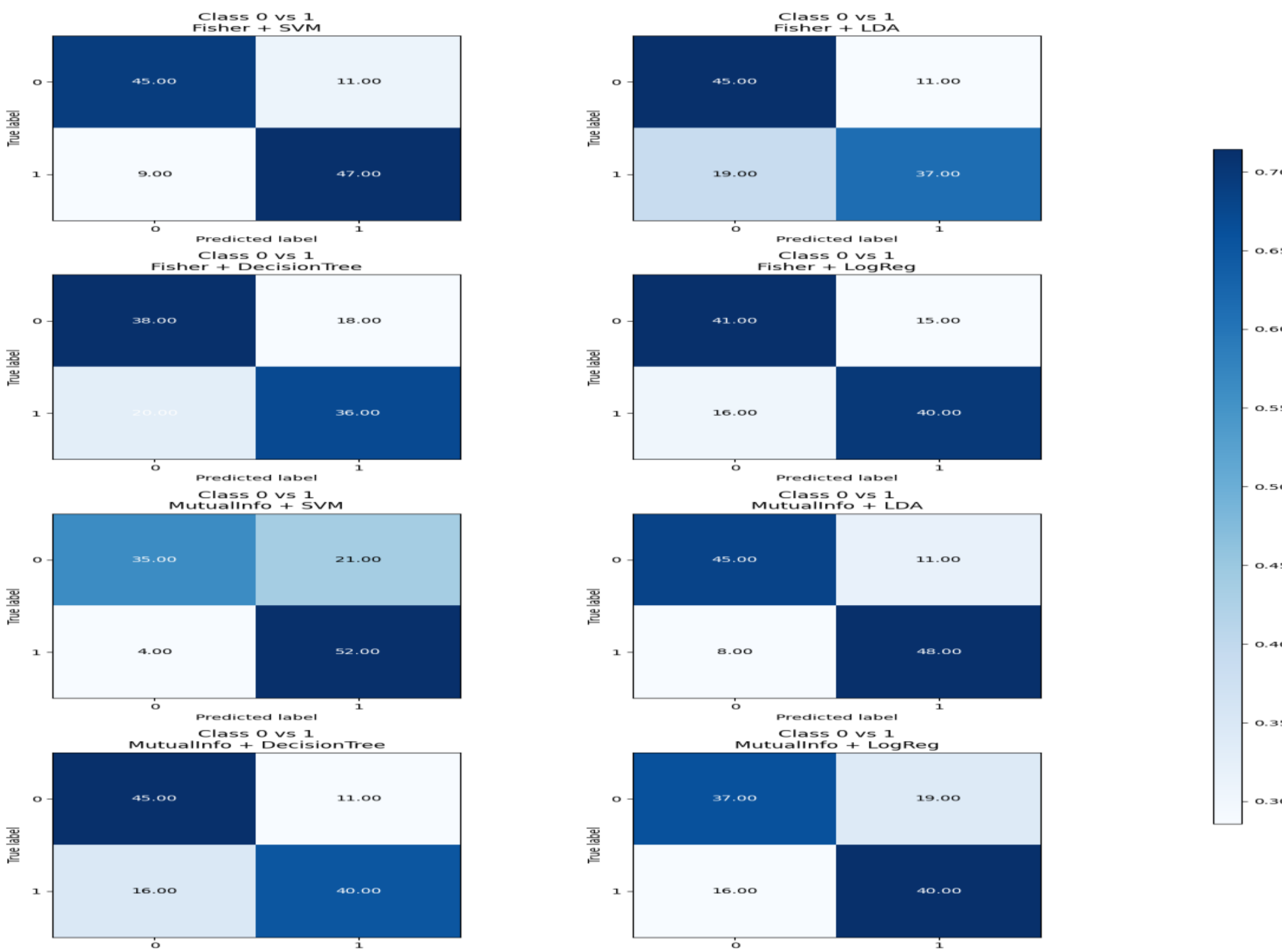

Figure 9.2 – Confusion matrices for Class 0 vs 1.

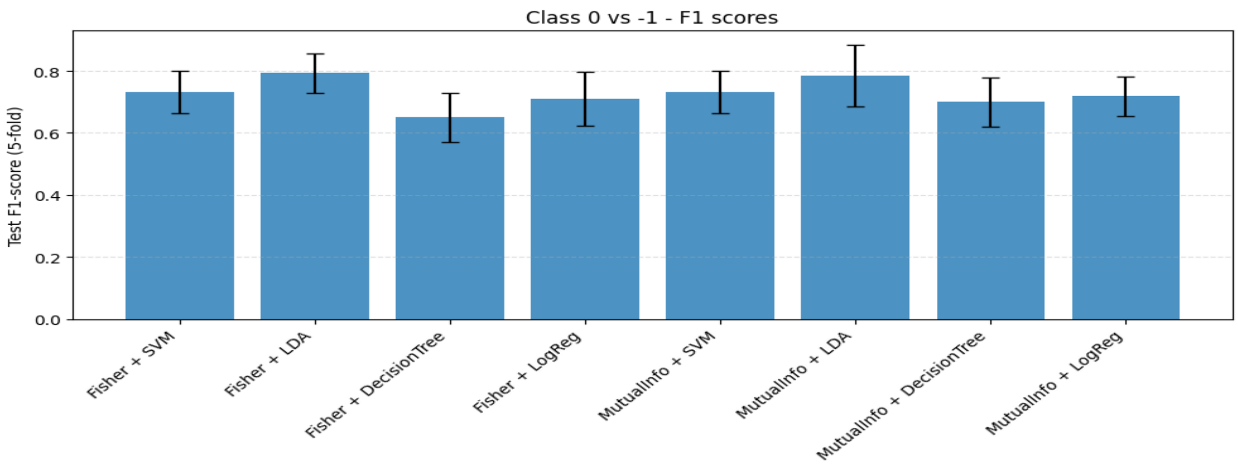

Figure 10.1 – F1-scores for binary classification (Class 0 vs -1).

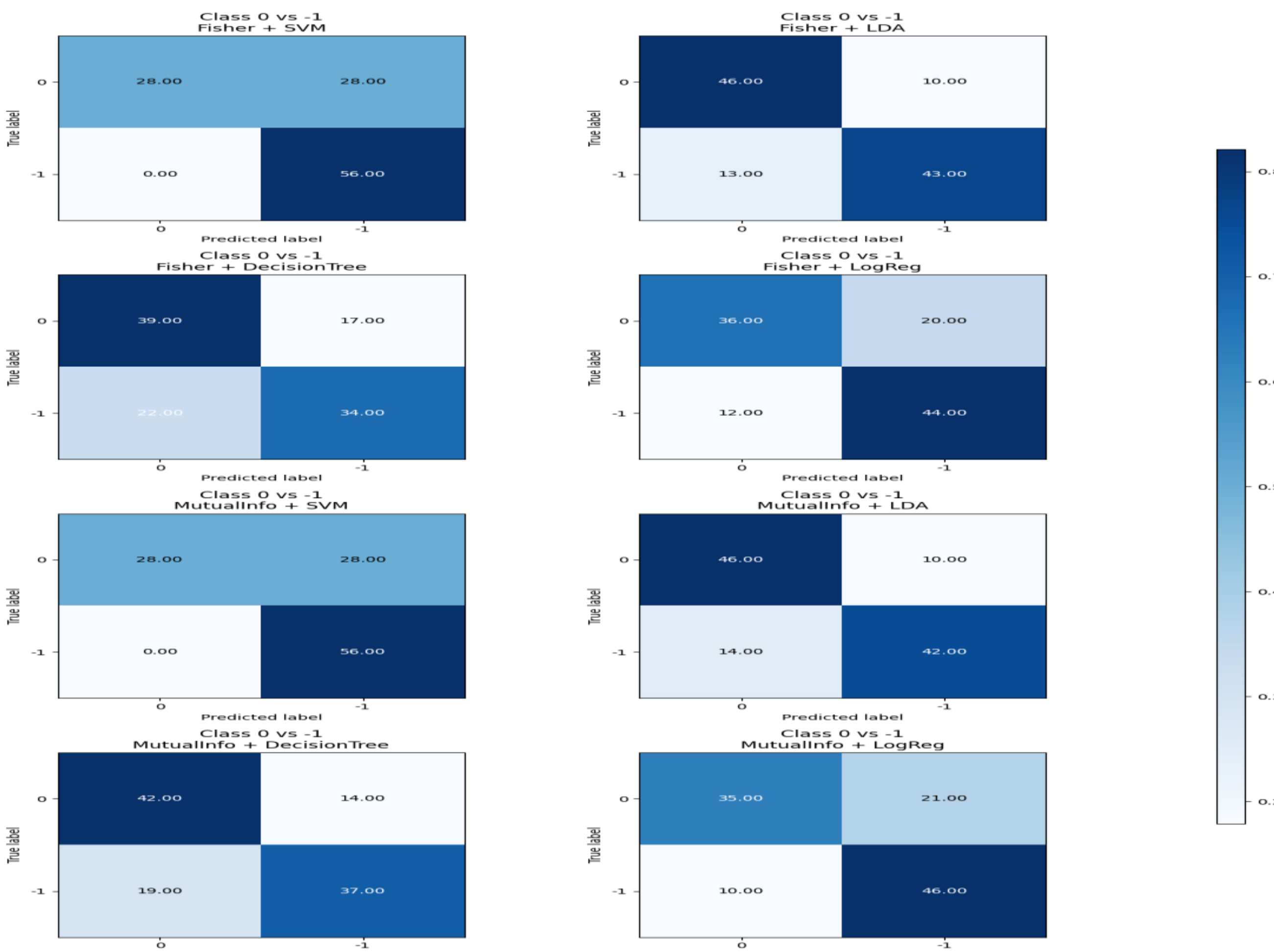

Figure 10.2 – Confusion matrices for Class 0 vs -1.

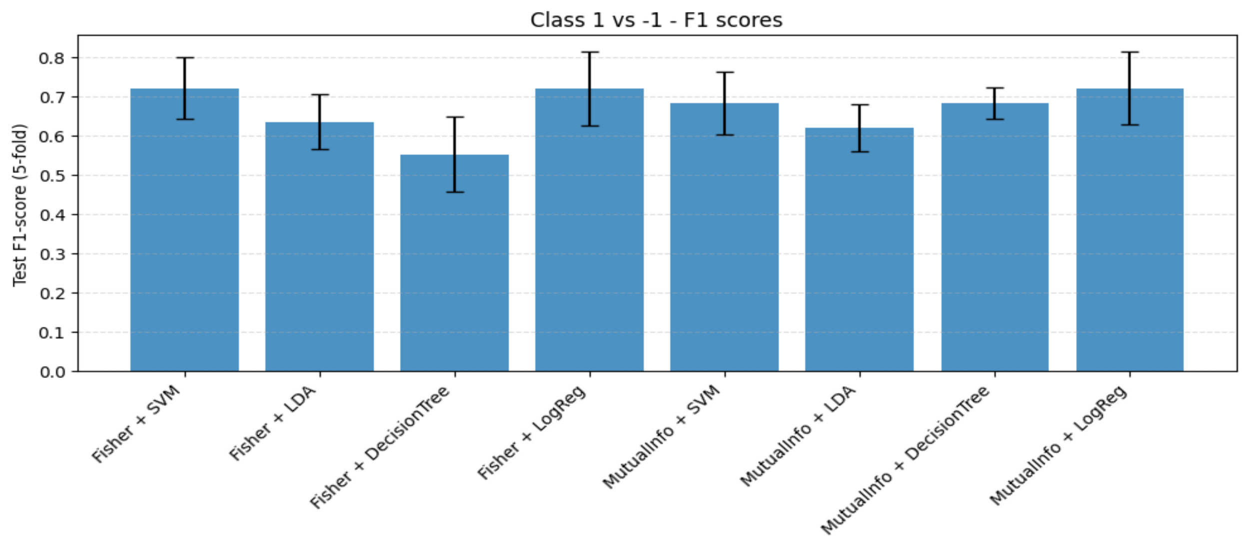

Figure 11.1 – F1-scores for binary classification (Clas -1 vs 1).

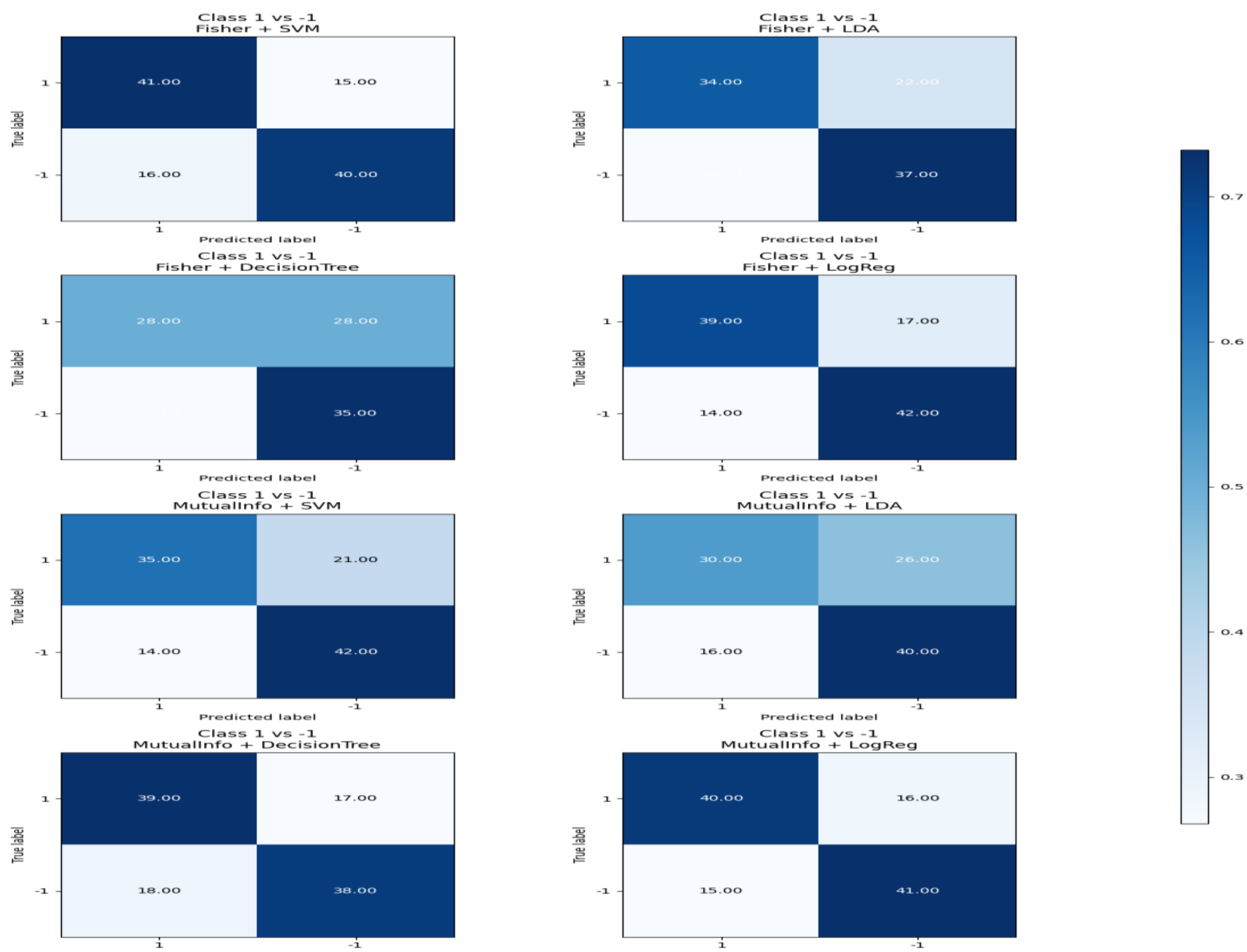

Figure 11.2 – Confusion matrices for Class -1 vs 1.

Discussion

This study examined the behavior of multiple classical machine learning classifiers across two distinct biomedical contexts: the WDBC breast-cancer dataset and a multi-class EEG signal classification task. Across both domains, several consistent patterns emerged regarding the influence of training set size, noise robustness, feature selection strategies, and the effectiveness of wavelet-based augmented features.

A key finding in the cancer dataset was the strong dependency of distance-based classifiers on the availability of representative training samples. KNN and Parzen Window showed substantial performance gains as the training set expanded, reflecting the importance of dense, well-populated neighborhoods for reliable local decision-making. In contrast, Naive Bayes and Nearest Mean were less sensitive to training set size, as their parametric assumptions allowed them to generalize reasonably well even from reduced data. Noise experiments further highlighted structural differences between the models: Naive Bayes degraded severely under high noise due to violations of its feature-distribution assumptions, whereas KNN and Parzen Window remained relatively stable at low noise levels but showed moderate degradation when noise displaced test samples away from their true local structure.

In the EEG classification task, the influence of feature selection and model architecture was more pronounced. Using the original 80 handcrafted features, SVM consistently outperformed the other classifiers under both Fisher and Mutual Information selection criteria, suggesting that the EEG feature space is highly nonlinear and benefits from the margin-maximizing properties of kernel methods. LDA performed moderately well, especially when Fisher scores were used, consistent with the linear separability assumptions embedded in shrinkage-based covariance estimation. Decision Tree models showed the largest variance and lowest mean performance, reflecting their sensitivity to feature noise and the relatively small dataset size. Logistic Regression delivered stable mid-range performance but was limited by the inherently nonlinear structure of the EEG classes.

A major improvement emerged when wavelet-based db4 features were appended to the original representation. Adding 20 wavelet features (10 per channel) increased the discriminability of temporal-frequency patterns associated with the EEG classes, leading to substantial improvements across nearly all classifiers. The improvement was most pronounced for SVM and Logistic Regression, indicating that the added features introduced richer linear and nonlinear separability. Fisher selection benefited particularly from the wavelet set, as wavelet energies and statistics often exhibit large between-class variance relative to within-class variance. Mutual Information selection also improved classification, but to a slightly lesser degree, as MI tends to favor features with strong marginal relevance even when redundant.

The binary EEG evaluations (0 vs 1, 0 vs –1, and 1 vs –1) further demonstrated that class separability varies significantly across pairs. For example, the 0 vs 1 pair showed high separability with both Fisher and MI selection, particularly for SVM and LDA. In contrast, the 1 vs –1 pair showed lower separability for several models, indicating that these two states share more overlapping EEG characteristics. Across all binary tasks, the wavelet-enhanced feature set provided consistent improvements, confirming its utility for sharpening the discriminatory structure of EEG representations.

Overall, the results emphasize several important conclusions for biomedical signal classification. First, model performance is shaped strongly by the alignment between classifier assumptions and underlying data structure. Second, noise robustness is not uniform across models and must be considered when designing practical diagnostic tools. Third, feature engineering remains a decisive factor in EEG-based systems, and time-frequency features extracted via wavelets contribute significant discriminative value. Finally, combining strong feature selection with expressive classifiers such as SVM offers a robust pathway for improving classification accuracy across heterogeneous biomedical datasets.

Conclusion

This study examined the behavior of several classical machine learning models across two biomedical classification tasks: breast cancer detection using the WDBC dataset and EEG-based mental state classification. The results showed that model performance depends strongly on data size, noise conditions, and feature representation. In the WDBC dataset, larger training sets consistently improved accuracy, with KNN and Parzen Window benefiting the most, while Naive Bayes remained highly sensitive to noise.

For the EEG dataset, SVM achieved the highest and most stable performance under both Fisher and Mutual Information feature selection. The addition of db4 wavelet-based features further improved results across all classifiers, confirming the value of time–frequency information in EEG classification. Binary EEG evaluations also revealed differences in class separability, with some class pairs being easier to distinguish than others.

Overall, the findings highlight the importance of appropriate feature extraction, robust model selection, and evaluation under realistic conditions. The study demonstrates that combining wavelet-based features with margin-based classifiers such as SVM provides a strong foundation for EEG classification tasks. Future work may investigate cross-subject generalization and advanced learning architectures to further enhance performance.

References

[1] S. R. e. al, A comparative study of machine learning techniques for breast cancer classification, vol. vol. 135, Expert Systems with Applications, 2019.

[2] H. A. e. al, "Using Machine Learning Algorithms for Breast Cancer Risk Prediction and Diagnosis," *Procedia Computer Science,* vol. Volume 83, pp. Pages 1064-1069, 2016.

[3] N.-D. M. e. al, "Real-Time On-Chip Machine-Learning-Based Wearable Behind-The-Ear Electroencephalogram Device for Emotion Recognition," *IEEE Access,* vol. 11, pp. 47258-47271, 2023.

[4] D. D. e. al, "EEG-based emotion recognition: Review of commercial EEG devices and machine learning techniques," *Journal of King Saud University - Computer and Information Sciences,* vol. 34, no. 7, pp. 4385-4401, 2022.

[5] G. H. e. al, "Learning from class-imbalanced data: Review of methods and applications," *Expert Systems with Applications,* vol. Volume 73, pp. 220-239, 2017.

[6] K. M. H. e. al, "Epileptic seizure detection in EEG using mutual information-based best individual feature selection," *Expert Systems with Applications,* vol. 193, p. 116414, 2022.

[7] M. N. e. al., "Machine learning algorithms for breast cancer prediction and diagnosis," *Procedia Computer Science.,* vol. 191, pp. 487-492, 2021.

[8] S. M. A. e. al, "Emotions Recognition Using EEG Signals: A Survey," *IEEE Transactions on Affective Computing,* vol. 10, no. 3, pp. 374 - 393, 2017.

[9] OpenAI, "ChatGPT: GPT-4 language model," [Online]. Available:, [Online]. Available: https://openai.com/chatgpt..